

South African Fuel Prices & Cost-of-Living Analysis

January 2021 – April 2024 | Gauteng, Limpopo, Western Cape & KwaZulu-Natal

Data Analytics Portfolio Project

SQL Data Cleaning, Analysis & Economic Insight Report

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Tools: MySQL (CTEs, window functions, generated columns)

Executive Summary

This project analyses a 160-record panel dataset tracking petrol and diesel prices alongside a basic food basket, transport fares, CPI and unemployment across four South African provinces (Gauteng, Limpopo, Western Cape, KwaZulu-Natal) from January 2021 to April 2024. The raw CSV export contained nine distinct data quality issues; a MySQL cleaning pipeline (CTEs, CASE logic, window functions and generated columns) was used to standardise, deduplicate and correct the data before running 22 analytical queries grouped into core, advanced and economic-insight tiers.

Headline finding : petrol prices rose 59.8% over the period (Gauteng: R15.09 → R24.12/L), but the basic food basket rose even faster — up 61.2% (R316 → R510) — and transport fares rose fastest of all, up 75.0%. Petrol price and CPI are strongly correlated ($r = 0.72$), confirming fuel as a key inflation driver in this data. Limpopo, this project's home province, consistently carries the highest petrol price of the four (an inland premium of ~18c/L over Gauteng) while also showing the lowest unemployment rate in the dataset.

Methodology

1. Data Source & Structure

The dataset comprises 160 cleaned records (40 months × 4 provinces) loaded into a `fuel_consumer_cleaned` table, with two SQL-generated columns computed at load time:

- **basic_basket_zar** — a weighted proxy for a basic food basket (bread + ¼×10kg maize meal + chicken + 4×milk + eggs).
- **petrol_affordability** — petrol price expressed as a percentage of the CPI index.

A companion `data_quality_log` table documents every cleaning decision, enabling a fully auditable pipeline.

2. Data Cleaning Process

Nine issues were identified and resolved in the raw CSV before analysis:

Issue	Field(s)	Rows	Action
Inconsistent date formats (3 variants)	month_year	3	Normalised to ISO date
Province abbreviation ("KZN")	province	1	Standardised to full name
Inconsistent city casing	city	2	INITCAP + TRIM applied
4 variant labels for one fuel type	fuel_type	4	Standardised to "ULP 95"
Redundant duplicate fuel-type row	fuel_type	1	Removed
Exact duplicate row (Apr-2023)	all columns	1	Removed via ROW_NUMBER()
Diesel outlier (R99.85, ~5x normal)	diesel_price	1	Corrected to R19.12
Missing petrol price (Limpopo, Jul-2022)	petrol_price	1	Imputed from Gauteng + premium
Prices stored as text	11 numeric cols	all	Explicit CAST to NUMERIC

A tenth issue surfaced during independent QA of this portfolio and is not in the original log: the `cpi_index` and `unemployment_pct` columns are transposed relative to their names. The column labelled `cpi_index` holds values in the 10–19.5 range (consistent with an unemployment rate) and varies by province, while the column labelled `unemployment_pct` holds a smoothly rising national value of 116–145 (consistent with a CPI index rebased near 100). All figures in this report use the corrected mapping, and this relabeling is flagged explicitly as a limitation below rather than silently changed in the source file.

3. Analytical Framework

Twenty-two SQL queries were organised into three tiers:

- **Core (A1–A8):** descriptive summaries — provincial averages, annual trends, food basket comparisons, cheapest/most expensive months, diesel-petrol gap.
- **Advanced (B1–B8):** year-over-year change, provincial price premiums, affordability index, rolling averages, running totals, monthly province rankings — built with window functions (LAG, RANK, running SUM).
- **Economic Insights (C1–C6):** before/after event comparison around the Jun-2022 price spike, Pearson correlation, fare elasticity, a composite economic-stress index, and a Western Cape vs Gauteng cost-of-living comparison — combining CTEs with statistical aggregation.

Key Insights

1. Petrol prices rose 60% in 3½ years, tracking a mid-2022 global shock

Gauteng petrol rose from R15.09/L (Jan-2021) to a series peak of R26.32/L (Jul-2022) before easing to R24.12/L by Apr-2024 — a net increase of R9.03, or 59.8%. The five most expensive months were all Jun–Sep 2022, coinciding with the Russia-Ukraine war's global oil-price shock flagged in the source notes; the five cheapest were all Jan–May 2021.

2. The food basket and transport fares outpaced fuel itself

While petrol rose 59.8%, the basic food basket rose 61.2% (R316.29 → R509.93 average) and the transport fare rose fastest of all at 75.0% (R19.99 → R34.99, Gauteng). This suggests second-order cost pass-through — transport and food margins absorbing fuel cost increases with a lag, then compounding — rather than fuel being the sole driver of living-cost pressure.

3. Fuel is a strong, not perfect, CPI driver

Petrol price correlates strongly with CPI nationally (Pearson $r = 0.72$) and in Gauteng specifically ($r = 0.72$), and moderately-strongly with the unemployment rate ($r = 0.69$). A composite economic-stress index (combining petrol price, unemployment and CPI) climbed every year in the dataset — from 17.5 in 2021 to 27.1 by 2024, a 55% increase — indicating compounding, not isolated, pressure on households.

4. Limpopo pays a consistent inland fuel premium — but has the lowest unemployment

Limpopo's petrol price sits ~18c/L above Gauteng in every month (range: 13c–21c), consistent with South Africa's inland fuel-zone differential. Notably, Gauteng, Western Cape and KwaZulu-Natal share an identical petrol price series in this dataset — a simplification worth flagging (see Limitations). Despite the fuel premium, Limpopo shows the lowest average unemployment rate of the four provinces (12.2%), against Western Cape's 16.2%, Gauteng's 14.2% and KwaZulu-Natal's 14.1%.

5. Western Cape carries the highest cost of living despite equal fuel pricing

Western Cape's average food basket (R410.17) runs 4.2% (~R16) above Gauteng's (R393.99), even though the two provinces track an identical petrol price. Limpopo's basket (R405.47) is also above Gauteng's despite Limpopo being the lower-income province by unemployment — implying food-basket cost, not fuel price, is the more discriminating cost-of-living variable between provinces in this dataset.

Summary Table — Provincial Comparison (period averages)

Province	Avg Petrol (R/L)	Avg Basket (R)	Avg Unempl. (%)
Limpopo	21.10	405.47	12.2
Gauteng	20.92	393.99	14.2
KwaZulu-Natal	20.92	387.17	14.1
Western Cape	20.92	410.17	16.2

Recommendations

- **1. Build a transport/food pass-through model.** Because fares (75%) and basket costs (61%) outpaced petrol itself (60%), a follow-up regression estimating the lag between a fuel move and its downstream pass-through into fares and food prices would give more actionable early-warning value than tracking petrol alone.
- **2. Investigate the flat pricing across Gauteng/Western Cape/KwaZulu-Natal.** In reality these provinces sit in different fuel zones; the identical series here likely reflects a simplified data-generation process rather than the true Wholesale/Retail zone differential. Any operational use of this pipeline on live data should validate the province-level fuel zone mapping before trusting the premium figures.
- **3. Track the economic-stress index quarterly, not just annually.** The composite index rose every year of the dataset (+55% cumulative); a quarterly cut around known shock events (e.g., the Jun-2022 spike) would show whether stress accumulates gradually or in step-changes, which matters for the timing of any planned intervention or budget adjustment.
- **4. Prioritise Western Cape and Limpopo in basket-affordability review.** Both provinces run above the Gauteng benchmark despite different unemployment profiles, so a single national assumption about affordability understates pressure in both — for different underlying reasons (cost-of-living vs. income).

Limitations

- The dataset is a compact, evenly-spaced synthetic panel (exactly 4 rows/month, no gaps beyond the one imputed cell) rather than a live survey export, so seasonality and noise are likely understated relative to real StatsSA data.
- The cpi_index / unemployment_pct column-name transposition (see Methodology) means any downstream system consuming this table by column name, rather than by validated range, would silently invert the two metrics.
- Petrol pricing is identical across three of the four provinces, which does not reflect South Africa's actual multi-zone wholesale/retail fuel pricing structure.
- Correlation values (e.g., $r = 0.72$ for petrol vs. CPI) describe association within this sample only and should not be read as a causal fuel→inflation estimate without controlling for other CPI components.

End of report.